Kevin (Shuo-Wen) Chang

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Google Scholar: https://scholar.google.com/citations?user=huNBqB0AAAAJ

PROFESSIONAL SUMMARY

Physical design and APR engineer with hands-on experience executing Innovus place-and-route flows across production designs from Apple, Qualcomm on advanced technology nodes. Deep expertise in placement, clock tree synthesis, routing, timing closure, and DRV debugging, combined with strong analytical skills developed through systematic QoR evaluation and root-cause analysis.

Proven ability to identify and resolve critical implementation issues — including 100+ customer-impacting bugs at BLUE/RED severity — through disciplined analysis of APR optimization behavior and timing/power degradation. Additional experience correlating production silicon behavior with design implementation characteristics.

- Digital Implementation & P&R: Innovus, ICC2 (equivalent APR tool), full flow execution — floorplan, placement, CTS, routing, timing closure, DRV fixing, QoR optimization
- Timing & Power Analysis: static timing analysis (STA), PPA evaluation, delay modeling, timing/power degradation analysis
- Programming & Scripting: Tcl, Perl, Python, Shell (EDA flow automation and data analysis)
- Physical Verification & Sign-off: DRV analysis, DRC awareness (Calibre), LEC concepts; familiarity with advanced-node sign-off flows
- Languages: Mandarin (Native), English (Professional), Japanese (JLPT N3, CEFR B1)

PROFESSIONAL EXPERIENCE

Qualcomm Semiconductor Corporation, Engineer on AI CAD Tool — Hsinchu, TW 08/2021 – Present

- Architected a scalable Multi-Agent AI platform utilizing the Model Context Protocol (MCP) to bridge the semantic gap in hardware design; unified natural language inputs, GB-scale EDA logs, and binary waveform data (FSDB) into an automated debugging ecosystem.
- Developed a production-grade Waveform & EDA Agent using Few-Shot Prompting and constrained JSON schemas, enabling an LLM to autonomously translate engineer intent into high-level Tcl command sequences that drive on-prem Verdi and Cadence Innovus APIs.
- Engineered a closed-loop Self-Correction mechanism that tracks background Linux subprocess stdout errors, allowing the AI to automatically debug and rerun its own generated scripts within 2–3 seconds, maintaining a 100% execution success rate.
- Designed strict enterprise security guardrails by isolating AWS IAM roles and sensitive credentials within local MCP servers, preventing LLM exposure to API keys and ensuring 100% corporate security compliance.
- Analyzed large-scale test datasets with NYCU lab to identify abnormal design behavior and systematic failure patterns related to DFT, MBIST, and ATPG diagnosis

Cadence Design Systems, APR Engine Engineer, Innovus R&D Dept. — Hsinchu, TW 02/2020 – 07/2021

- Executed Innovus APR flows hands-on across production designs from Apple, Qualcomm, on advanced process nodes — covering placement (place_opt_design), clock tree synthesis (ccopt_design), routing (routeDesign), and optimization (optDesign) — as the sole optimization-engine engineer in Taiwan

- Analyzed PPA (power, performance, area) and runtime behavior across diverse advanced-node designs, identifying implementation bottlenecks and driving improvements in timing closure and QoR
- Conducted detailed analysis of Design Rule Violations (DRVs) across floorplans, netlists, and layout data; deeply analyzed timing and power degradation caused by APR optimization algorithms
- Verified and evaluated machine-learning–assisted P&R optimization features within the GigaOpt engine, focusing on cell delay model QoR, slew/delay prediction accuracy, and PPA impact
- Collaborated closely with R&D teams to debug complex tool behavior and validate fixes for customer-reported issues on advanced-node (N4) technologies
- Slashed Innovus P&R turnaround time by 63% (from 4 hours to 1.5 hours) with zero regression by systematically profiling the POD pipeline, strategically optimization setup target slacks, and fine-tuning area reclamation constraints.
- Authored and scaled technical training methodologies across internal CAD pipelines based on successful timing optimization frameworks, elevating team-wide tool efficiency and debugging velocity.

Interuniversity Microelectronics Centre (IMEC), **International Scholar** — IMEC HQ, Belgium 09/2018 – 02/2019

- Participated in an international research project developing data-driven automation systems for bio-experiment workflows.
- Implemented software modules based on real experimental data generated by the doctoral scientists.
- Co-authored a high-impact **publication** in Frontiers in Neuroscience (2019), achieving 100+ citations per google scholar.

Mediatek Singapore Pte.Ltd, **Machine Learning Algorithm Engineer Intern**— **Singapore** 08/2019 – 11/2019

- Surveyed and benchmarked state-of-the-art Neural Machine Translation (NMT) model compression techniques for mobile device execution.
- Applied several model compression techniques to models through attention complexity visualization and head pruning.

INTERNATIONAL & CROSS-SITE EXPERIENCE

- Research internship at IMEC (Belgium) with European engineers and scientists
- Exchange student at KU Leuven (Belgium) — coursework and collaboration entirely in English
- Engineering internship at MediaTek (Singapore)
- Comfortable working across time zones, cultures, and engineering organizations

EDUCATION

National Taiwan University, MSc in Electrical Engineering (Computer Science gp.) — Taiwan 08/2017 – 01/2020

- GPA: 4.23 / 4.30, Rank: **8** / 91 | Coursework: Artificial Neural Networks (A+), Big Data Systems (A+)

Katholieke Universiteit Leuven, Exchange Student — Belgium 08/2018 – 02/2019

National Taiwan Normal University, BEng in Mechatronic Engineering — Taiwan 09/2013 – 06/2017

- GPA: 4.11 / 4.30, Rank: **1** / 59. Awarded with 4-year full government scholarship by Ministry of Education, Taiwan.

AWARDS

- **NTU Outstanding Student Award** (only 13 receivers over the universities in 2019) — 2019
- **NTU Outstanding Achievement Award** on behalf of EECS (only 1 receiver in college EECS in 2018) — 2018
- **IEEE VTS Best Paper Award** (2023), **International** — Presented 2024 Nov, 3rd author of the research paper
- **Microsoft Imagine Cup Taiwan**, **National** — 2nd Place in 2018 TW national competition
- Future Finance & Healthcare Hackathon, **National** — 1) 3rd Place as well as 2) People's Choice Award with top1 voted